

# ASHIRWAD KUMAR

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## Education

### B.Tech in Computer Science & Engineering

Graphic Era Deemed to be University, Dehradun

Class 12th – 86.4%

CGPA: 8.5/10

2023 – 2027

2021 – 2022

## Projects

### Club Activity Management System

Source Code

- Built a full-stack Club Activity Management System using **Python Flask, MySQL, and Jinja2**, enabling students to explore clubs and events through a dynamic, database-driven interface.
- Designed a relational MySQL schema with foreign key constraints supporting **100+ clubs** and automated activity status updates based on event timestamps.
- Built an admin dashboard with **dynamic CRUD operations**, eliminating frontend code changes when managing clubs and activities.
- Implemented **secure user authentication, session management, and REST-style routing** for efficient and safe navigation.

### E-Commerce Platform with Customer Segmentation

Live Demo

- Built scalable **REST APIs** using Flask with JWT-based authentication and transactional workflow management.
- Optimized an **ACID-compliant relational database** with indexed queries supporting multiple concurrent users.
- Applied **K-Means Clustering** over **10,000+ records** to generate insights on seasonal trends and customer purchasing patterns.
- Visualized sales trends and user behavior via interactive dashboards with a fully **responsive frontend**.

### Deep Packet Inspection (DPI) Engine

Source Code

- Built a **C++ DPI Engine** that parses raw network packets across **Ethernet/IP/TCP** layers and extracts **TLS SNI fields** from HTTPS handshakes to classify and block applications like YouTube and Facebook.
- Implemented **stateful Five-Tuple flow tracking** via **HashMap**, enabling persistent block decisions across all packets of a connection without redundant re-inspection.
- Architected a **multi-threaded pipeline** (Reader → Load Balancer → Fast Path → Output Writer) using **consistent hashing** for flow affinity and **thread-safe queues with mutex/condition variables**.
- Engineered a **rule-based policy engine** supporting IP blacklists, app-level filtering, and domain keyword matching, with per-application traffic classification reports.

### Intelligent Relocation and POI Optimization System

Source Code

- Built an Intelligent Relocation Platform using **React, Node.js, and PostgreSQL/PostGIS** to identify optimal living regions based on user-defined POI priorities such as hospitals and schools.
- Applied **DBSCAN clustering algorithm** to analyze geospatial POI data, automatically identifying high-density regions that best match user preferences for smarter relocation recommendations.
- Integrated **PostGIS** with radius-based spatial search, spatial indexing, and optimized queries to efficiently store and retrieve geospatial data at scale.
- Built interactive **geospatial visualizations using Leaflet.js**, enabling real-time dynamic region plotting and side-by-side regional comparisons on an interactive map.

## Technical Skills

**Languages:** Python, Java, C++, JavaScript

**Backend:** Node.js, Express.js, Flask, REST APIs, JWT Authentication, MVC

**Frontend:** HTML, CSS, Tailwind CSS, React

**Databases:** MySQL, PostgreSQL, MongoDB

**Developer Tools:** Git, GitHub, Postman, VS Code, Supabase

## Algorithmic Competitions & Achievements

- **3-Star Coder (Max Rating: 1700)** on [CodeChef](#).
- **Max Rating: 2126** on [LeetCode](#) with **1000+ problems solved**.
- **Ranked 514** among **28,000+** participants in LeetCode Weekly Contest 475.
- **Ranked 145** among **3,000+** participants in CodeChef Starters 220.
- Certification in **AWS - EC2 and VPC**.